MSCC on Raspberry Pi

Install in plain English

For: Raspberry Pi 4 or 5, 64-bit Raspberry Pi OS (desktop recommended).

Goal: Install MSCC, configure it, and start/stop the servers — mostly point-and-click after package install.

Current package examples (use the real filenames you were given):

Package	Example filename	Order
PortAudio	mscc-portaudio_19.8.2_arm64.deb	1st
Main stack	mscc_1.0.27_arm64.deb	2nd
Setup wizard	mscc-init-gui_1.0.10_all.deb	3rd (recommended)

Big picture (read this once)

Where	What runs
Raspberry Pi	Three servers (radio + audio stack) and setup tools
Your PC	MSCC client (screen, knobs, spectrum) — how you operate

The Pi does not need the MSCC client window. That is normal: headless servers on the Pi, GUI on the PC.

```
PC (MSCC client)  ← network →  Pi (ms-sdr + sdrcore-recv + sdrcore-trans)  ← USB → Proficio
```

Two audio worlds on the Pi:

Path	What you hear / use	Devices
Operator (phones)	Headphones / mic at the Pi	Real sound card — see Operator audio hardware below. Any sample rate is OK (servers adapt).
Digital (WSJT etc.)	Digi app audio on the Pi	VirtualA / VirtualB (software cables — not a physical cable)

Operator audio hardware (important on Pi 4)

The Proficio radio always uses USB audio for I/Q. Operator phones and mic should preferably use a different kind of path so the Pi is not running two busy USB sound devices at once.

Operator phones / mic	Pi 4	Pi 5	Notes
I²S audio HAT (e.g. AudioInjector, Codec Zero, HiFiBerry with ADC+DAC)	Recommended	Recommended	Proven, stable approach

Pi headphone jack / onboard audio	OK for listen	OK	Limited mic options
Digital only (VirtualA / VirtualB)	Recommended for digi	Recommended	No second physical sound card
Second USB headset or USB sound dongle (with Proficio also on USB)	Not recommended	Often OK	On Pi 4, under load this can interrupt USB audio; spectrum and phones may freeze while servers still look "running"

In short: On Raspberry Pi 4, choose an I²S HAT or the Pi jack for operator audio, or work digi via VirtualA/B. Reserve a USB headset plus Proficio mainly for Pi 5, or accept that dual USB audio is less reliable on Pi 4.

This is a supported configuration choice, not a defective install: Pi 4 is fully supported when operator audio is not a second USB sound device.

What you need

1. Pi 4/5 with Raspberry Pi OS desktop (menu + Terminal).
2. The three .deb files on the Pi (USB stick, Downloads folder, etc.).
3. Optional helper: install-mscc.sh (same folder as the main .deb).
4. Network on the Pi the first time (so apt can pull libraries).
5. Multus / Proficio powered on when you configure (recommended).
6. PC and Pi on the same network (or as your site requires) for the MSCC client.
7. Operator audio plan (see table above) — especially on Pi 4, plan an I²S HAT or digi path rather than a USB headset next to the Proficio.

Important: use Terminal from the Desktop

Package install still needs a short command line. Do this on the Pi:

8. Log into the Raspberry Pi desktop (monitor or usual desktop session).
9. Open Terminal from the desktop menu.
10. Run the install commands in that Terminal.

You only need enough CLI to type a few lines and enter your password. After install, use the menu:

Sound & Video → MSCC Init, MSCC Start, MSCC Stop

(If the menu layout changes, search the applications menu for those three names.)

Install steps

1) Install PortAudio for MSCC (required first)

In Desktop Terminal, go to the folder with the .deb files:

```
cd ~/Downloads
# or:  cd /media/pi/YOUR_USB_NAME
```

MSCC digi (VirtualA / VirtualB) needs a special PortAudio build (Pulse + ALSA). The stock Raspberry Pi OS library alone is often not enough.

```
sudo apt install -y ./mscc-portaudio_19.8.2_arm64.deb
```

(Use the real filename if the version number differs.)

This installs under /usr/local/lib. MSCC servers are built to use that copy.

Quick check (recommended):

```
ldconfig -p | grep portaudio
# good: a line with /usr/local/lib/libportaudio.so.2
```

After the main package is installed, also check:

```
ldd $HOME/mscc/sdrcore-recv | grep portaudio
# good: /usr/local/lib/...
# bad:  only /lib/aarch64-linux-gnu/...  → digi will often fail
```

Do not rely on old $HOME/portaudio-install + .bashrc exports for a normal install. Use this package.

2) Install the main MSCC package

Helper script (if you have it):

```
chmod +x install-mscc.sh
./install-mscc.sh
```

Or install the .deb directly:

```
sudo apt update
sudo apt install -y ./mscc_1.0.27_arm64.deb
```

(Use the real filename if the version differs.)

- Enter your password when asked.
- If it asks to continue / stop old servers → type y when ready.
- Install does not run the setup wizard for you — that is the next step.

You may see a technical "Notice" about _apt. If the install finished and printed "MSCC installed" / "done", ignore that notice.

This package installs:

Piece	What it is
Servers	ms-sdr, sdrcore-recv, sdrcore-trans → $HOME/mscc/
Start/stop	mscc command and desktop MSCC Start / MSCC Stop
Digi audio	mscc-virtual-audio creates VirtualA / VirtualB
Config seed	$HOME/.local/mscc/ only if empty (never overwrites your files later)
CAT helper	tty0tty module → often /dev/tnt0 (after build / reboot)

3) Install the GUI setup wizard (recommended)

```
sudo apt install -y ./mscc-init-gui_1.0.10_all.deb
```

(Use the real filename if the version differs.)

Adds menu MSCC Init. Needs desktop packages such as python3-tk and python3-pyaudio (usually installed automatically).

4) Log out / in once (first install)

If the install added groups (dialout, plugdev, audio) or audio services: log out and log back in, or reboot once.

Then check digi sinks:

```
pactl list short sinks | grep Virtual
```

If empty:

```
systemctl --user enable --now mscc-virtual-audio
# or run once:  mscc-virtual-audio
```

Role	Device name
Digital speaker (recv digi out)	VirtualA
Digital mic (trans digi in)	VirtualB.monitor
WSJT Input (hear radio)	VirtualA.monitor
WSJT Output (TX audio)	VirtualB

5) Configure this Pi (required once)

Preferred (point and click):

11. Applications menu → Sound & Video (typical).
12. Click MSCC Init.

The wizard:

- Offers to stop servers if they are already running (safe if you refuse and exit).
- Keyer, CAT port / PTT pin.
- Operator speaker and microphone — pick your phones / HAT. Any sample rate is OK (48 kHz, 96 kHz, etc.). Do not pick Proficio / Multus I/Q for headphones or mic. On Pi 4, prefer an I²S HAT or Pi headphones over a USB headset (see Operator audio hardware).
- Digi is fixed: VirtualA / VirtualB.monitor (not a menu choice).
- After save: Yes/No to start the MSCC servers.
- Sets operator hardware mixer levels toward full open (100%) so the MSCC client Volume can control loudness.

CLI alternative (SSH or no desktop):

```
mscc-init
```

Config files live under:

```
$HOME/.local/mscc/
```

Upgrades do not overwrite existing config.

6) Start and stop the servers (everyday)

Preferred (point and click):

Menu (usually Sound & Video)	What it does
MSCC Start	Starts digi sinks (best effort) + starts the three servers. Also re-applies operator speaker/mic ALSA levels to full (so phones stay loud after reboot).
MSCC Stop	Stops the three servers.
MSCC Init	Configure again (stops servers first if needed).

You may get a desktop notification if libnotify-bin is installed.

CLI alternative:

```
mscc start
mscc status
mscc stop
mscc restart
```

Then use the MSCC client on your PC to operate the radio (Pi IP / hostname as configured).

Operator phones vs digi (what to expect)

Phones (operator path)

- Proficio radio I/Q is always 96 kHz on the wire.
- Your operator device can be 96 kHz or not (e.g. 48 kHz). The servers handle the difference.
- On Pi 4, best results with an I²S HAT or Pi jack — not a second USB sound card (see Operator audio hardware).
- Volume after reboot: run MSCC Start (or mscc start). Levels are re-applied automatically.
- If still quiet: re-run MSCC Init, or raise levels once with the desktop volume control / alsamixer for that sound card.
- Day-to-day loudness: use the MSCC client Volume control.

Digi (WSJT etc.)

Typical WSJT-X on the Pi:

WSJT setting	Value
Input	VirtualA.monitor
Output	VirtualB
CAT	As set in MSCC Init (often /dev/tnt0)

Digi uses VirtualA / VirtualB (software). That does not add a second USB sound device, so it is a good fit for Pi 4 as well as Pi 5.

Digi levels use Pulse/PipeWire, not the same as phone alsamixer.

If WSJT's receive bar is weak while the MSCC client is already loud:

```
pactl set-sink-volume VirtualA 100%
pactl set-sink-mute VirtualA 0
```

(That raises the digi path into WSJT. Then trim with MSCC client Volume if needed.)

WSJT-X and CPU (especially Pi 4): MSCC already uses real CPU for the radio. FT8 with many decoder threads can load the Pi heavily. If the desktop feels sluggish or decodes fall behind, lower WSJT's thread count (or use single-thread decode). A Pi 5 has more headroom for MSCC + multi-thread FT8 together.

Virtual sinks should return at desktop login via the mscc-virtual-audio user service. If digi vanishes after reboot:

```
systemctl --user enable --now mscc-virtual-audio
```

Did it work?

```
mscc status
ls /dev/tnt0 /dev/tnt1
pactl list short sinks | grep Virtual
ldd $HOME/mscc/sdrcore-recv | grep portaudio
```

Check	Good sign
mscc status	sdrcore-recv, sdrcore-trans, and ms-sdr show running
/dev/tnt0	Exists (CAT / PTT when tty0tty built)
pactl … Virtual	VirtualA / VirtualB present
ldd … portaudio	/usr/local/lib
PC client	Connects to this Pi

Logs (if something fails):

- Often: ~/sdrcore-recv.log, ~/sdrcore-trans.log, ~/ms-sdr.log
- Or under: $HOME/.local/mscc/
- Start script text: $HOME/mscc/logs/

Useful digi / audio lines:

```
grep -E "VirtualA|DUAL STREAM|ALSA card|portaudio|FAILED" ~/sdrcore-recv.log | tail -30
```

Everyday use (after first install)

13. Power on Pi and radio.
14. Log into the Pi desktop (so Pulse/PipeWire and Virtual* can start).
15. Menu → Sound & Video → MSCC Start (or Terminal: mscc start).
16. Use MSCC on the PC as usual.
17. When done: MSCC Stop (optional if you leave the Pi running).

Re-run MSCC Init only if you change phones, HAT, or CAT wiring.

If something fails

Symptom	What to try
mscc: command not found	New Desktop Terminal, or log out/in
No menu entries	Confirm packages installed; look under Sound & Video; log out/in

<table>
<tr><td>mscc-init-gui missing</td><td>Install mscc-init-gui_*.deb; need desktop + python3-tk / python3-pyaudio</td></tr>
<tr><td>No /dev/tnt0</td><td>Reboot once; install headers: sudo apt install -y linux-headers-$(uname -r)</td></tr>
<tr><td>No VirtualA/B</td><td>mscc-virtual-audio or systemctl --user enable --now mscc-virtual-audio</td></tr>
<tr><td>Digi devices not listed in init / WSJT</td><td>Install mscc-portaudio first; check ldd → /usr/local</td></tr>
<tr><td>Wrong PortAudio (ldd shows only /lib/...)</td><td>Install mscc-portaudio; use mscc 1.0.23+</td></tr>
<tr><td>Quiet phones after reboot</td><td>MSCC Start (re-applies levels); re-run MSCC Init if needed</td></tr>
<tr><td>Quiet digi / WSJT RX weak</td><td>pactl set-sink-volume VirtualA 100%; WSJT Input = VirtualA.monitor</td></tr>
<tr><td>Audio wrong device</td><td>Re-run MSCC Init. Digi stays VirtualA / VirtualB.monitor</td></tr>
<tr><td>Spectrum / phones freeze; mscc status still running</td><td>Often USB audio dropped under load. On Pi 4, avoid USB headset + Proficio. Use an I^{2}S HAT or digi path. Then mscc stop / mscc start and reconnect the client. Check: dmesg -T | grep -iE 'disconnect|usb'</td></tr>
<tr><td>WSJT makes the Pi very busy / late decodes</td><td>Lower WSJT decoder threads on Pi 4; prefer digi VirtualA/B; consider Pi 5 for heavy multi-thread FT8 + MSCC</td></tr>
<tr><td>Permission denied on serial/tnt</td><td>Log out/in or reboot (groups dialout / plugdev)</td></tr>
<tr><td>Servers won’t start from menu</td><td>Terminal: mscc start and read the error text</td></tr>
<tr><td>Client won’t connect</td><td>Same network; firewall; correct Pi IP; mscc status</td></tr>
<tr><td>AudioInjector HAT silent after first install</td><td>Reboot once if the package added the audio overlay</td></tr>
</table>

This stack is only for Raspberry Pi OS 64-bit on Pi 4/5. Other Linux systems are not supported.

What the three servers are (optional reading)

Program	Job
sdrcore-recv	Receive: radio I/Q in, audio to phones and digi out
sdrcore-trans	Transmit: mic / digi in, radio I/Q out
ms-sdr	Control hub: talks to Proficio and to the PC client

You do not start them individually for normal use — use MSCC Start / mscc start.

One-line cheat sheet

```
Desktop Terminal (order matters):
  1. sudo apt install -y ./mscc-portaudio_*_arm64.deb
  2. sudo apt install -y ./mscc_*_arm64.deb
  3. sudo apt install -y ./mscc-init-gui_*_all.deb
  (log out/in if first time)
  Optional checks:
     ldconfig -p | grep portaudio
     ldd $HOME/mscc/sdrcore-recv | grep portaudio
```

```
     pactl list short sinks | grep Virtual

Menu → Sound & Video:
  MSCC Init  →  configure once  →  Yes to start (or MSCC Start later)
  MSCC Start / MSCC Stop day to day

PC: run MSCC client → connect to Pi

WSJT (on Pi): Input = VirtualA.monitor   Output = VirtualB

Pi 4 operator audio: I²S HAT or Pi jack preferred (not USB headset + Proficio)
Pi 4 + FT8: keep WSJT thread count modest
```

That’s it.

For developers / handoff

See README.md (package build) and resume.md (status, audio/pan contracts, open items). Do not put long engineering notes in this operator card — keep this file short and one path.